

Teratogenic effects of HESA-A, a natural anticancer product from Iran, in mice

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Abstract

HESA-A is a natural product containing herbal and marine animal substances, which has been patented in Iran. It has shown antioxidant, cytotoxic and anticancer effects. The aim of this study was to evaluate the teratogenic effects of HESA-A in mice. HESA-A (50, 100, 200, 400 and 800 mg/kg) was administered orally to pregnant mice on days 6 to 14 of gestation. Mouse reproductive developmental toxicity study was performed according to the ICH guideline. Embryos from the treated dam were sectioned and studied for external morphological abnormalities and skeletal malformations. HESA-A at two doses (400 and 800 mg/kg) slowed weight gain of pregnant mice. These two doses of HESA-A led to reduction in uterus weight (17% and 20% for the 400 and 800 mg/kg doses, respectively), increase in post-implantation resorption (150% and 200%, respectively), reduction in fetus weight (22% and 32%, respectively) and crown-rump length (15% and 19%, respectively). HESA-A at 400 and 800 mg/kg doses caused mild external and skeletal malformation significantly higher than the normal saline group. However, higher doses caused embryo malformations such as short limbs, spinal abnormalities, dermal cysts, microphthalmia and cleft palate. According to this study, only higher doses of HESA-A, which are 20 to 40 times higher than the usual therapeutic doses based on body surface conversion, may cause embryonic toxicity. This provides a reasonable safety margin for the use of HESA-A in pregnancy. Mechanisms of these abnormalities are not clear and need to be determined.

Keywords

HESA-A, embryotoxicity, mice, developmental toxicity

Introduction

HESA-A is an active natural compound with herbal and marine origin, which has been patented in Iran for its biological properties and has recently been permitted for mass production by the Islamic Republic of Iran's Ministry of Health and Medical Education (registration number: D-5-6638; dated on 14-06-2005). HESA-A contains mainly concentrated components from the marine species *Penaeus semisulcatus* and the plant species *Cuminum cyminum* and *Apium graveolens*. HESA-A has been shown in numerous animal and human studies to have strong anticancer properties with little impact on normal cells. In one study, this compound manifested selective concentration-dependent effects, comparable with those of doxorubicin, against the growth of Hela, HepGII and MDA-MD-468 malignant cells without significant impacts on normal cells.¹ In another in vivo study in rabbits, daily doses of 350 mg/kg of HESA-A

caused shrinkage of osteosarcoma tumors and resulted in total disappearance of tumor in 10 weeks. All the

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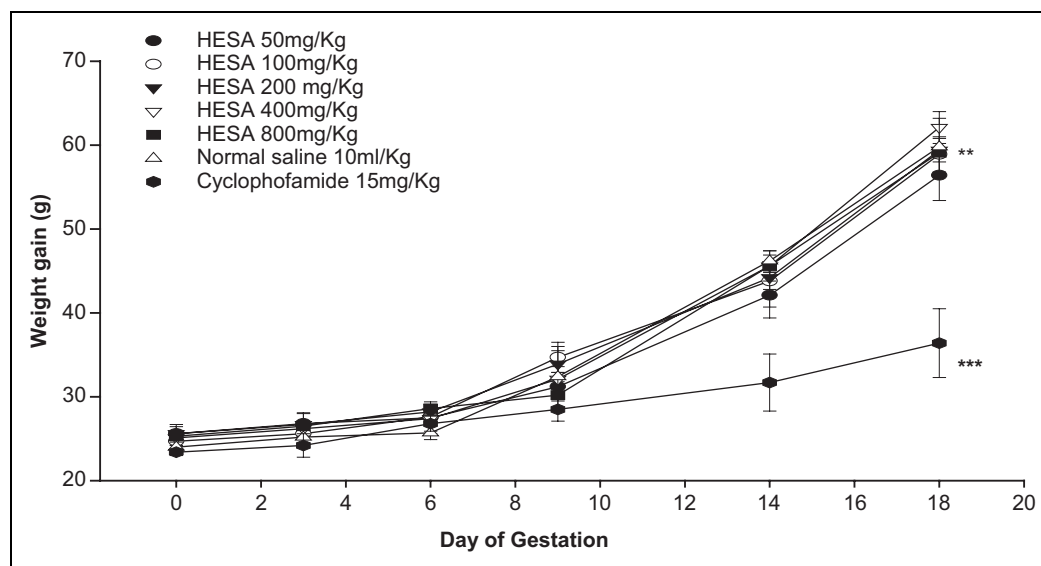


Figure 1. Weight changes in pregnant mice receiving HESA-A in different doses from day 1 through 20 of gestation, compared to normal saline and cyclophosphamide.

rabbits in the study survived, whereas those in the control group expired during the same period. Pathological studies confirmed the effects of HESA-A². In a double-blind clinical trial, 24 patients suffering from breast cancer with choroidal metastasis were treated with 50 mg/kg/day HESA-A and the findings indicated significant improvement of vision and pain toleration in patients compared with the control group.³ In a pilot study, 3 weeks of treatment with 50 mg/kg HESA-A was observed to manifest significant reduction in intensity and number of attacks in 38 patients with multiple sclerosis (unpublished data). Moreover, hepatoprotective effects of two doses of HESA-A (125 and 250 mg/kg) against hepatotoxicity caused by thioacetamide were shown in rabbits.⁴

Toxicological studies have shown that acute toxic effects of HESA-A present in values higher than 10 g/kg in mice, yielding an LD₅₀ of 16 g/kg in mice and 18g/kg in rats. At LD₅₀ doses, clinical symptoms included lethargy, vomiting, diarrhea and seizure. Pathological assessment detected mild lymphocyte infiltration in livers of few of the treated mice. Based on this and complementary pathological, biochemical and hematological studies, it has been concluded that HESA-A is practically non-toxic for mice and rats.⁵ Assessments of HESA-A mutagenicity using Ames test indicate that the number of revertant colonies is not significantly different from negative control and therefore this compound is mutagenically neutral.⁶

HESA-A analysis show that the compound is composed of 50% inorganic substance, 45% organic substance (aminoanthraquinone) and 5% water. The inorganic fraction includes calcium carbonate, magnesium phosphate and sulfate, potassium and sodium, also metals such as aluminum, cobalt, potassium, chrome, iron, zinc, brome and strontium with high concentrations.⁷ Considering the composing materials, there is a risk of teratogenicity.

It is well-known that many anticancer agents have teratogenic and toxic effects on fetuses, since embryonic cells have high rates of proliferation and due to the nonspecific effect of these compounds on embryonic and malignant cells, teratogenicity poses as a problem. This study was conducted to evaluate the potential of HESA-A for teratogenicity and determine its toxic dose in mouse fetus.

Method

HESA-A powder (Dr Ahmadi Institute, Isfahan, Iran) was dissolved for an hour in normal saline acidified with HCl, and the resulting solution was treated with NaOH to yield a final pH of 7.4.¹ The solution was filtrated and administered orally to mice on the sixth through fourteenth days of gestation in 50, 100, 200, 400 and 800 mg/kg doses.⁸ Normal saline and cyclophosphamide (15 mg/kg) were used as negative and positive controls, respectively.

Table 1. Means of variables in study in pregnant mice receiving different doses of HESA-A from day 6 through 14 of gestation

	HESA-A (mg/kg)					Cyclophosphamide (15 mg/kg)	
	50	100	200	400	800		Normal saline
No of pregnant mice	10	10	10	10	10	14	14
Weight of uterus/pregnant mouse (g) (mean $\pm$ SD)	17.49 $\pm$ 1.1	16.41 $\pm$ 1.3	15.27 $\pm$ 2.1	14.9 $\pm$ 1.2 ^a	14.4 $\pm$ 1.4 ^a	16.3 $\pm$ 1.8 ^b	18 $\pm$ 1.4
No of resections/pregnant mouse	2	2	3	5 ^a	6 ^a	5 ^a	2
No of fetuses/pregnant mouse	12	11	11	7 ^a	8 ^a	7 ^a	14
Total No of fetuses	120	110	110	70 ^a	80 ^a	98 ^b	196
Viable fetuses/pregnant mouse	11	10	9 ^a	6 ^c	6 ^c	6 ^c	13
Weight of fetuses (mg) (mean $\pm$ SD)	1344 $\pm$ 83	1329 $\pm$ 111	1273 $\pm$ 98	1187 $\pm$ 68 ^a	1035 $\pm$ 89 ^a	768 $\pm$ 45 ^c	1528 $\pm$ 121
Crown-rump length of fetuses (mm) (mean $\pm$ SD)	13.4 $\pm$ 0.9	12.7 $\pm$ 1.1	12.1 $\pm$ 0.7	11.9 $\pm$ 0.6 ^b	11.4 $\pm$ 0.6 ^b	6.9 $\pm$ 0.3 ^c	14.1 $\pm$ 0.8

^a $p < 0.01$.

^b $p < 0.05$.

^c $p < 0.001$.

Table 2. External abnormalities in fetuses receiving HESA-A in different doses

External findings	HESA-A (mg/kg)					Cyclophosphamide (15 mg/kg)	Normal saline
	50	100	200	400	800		
No fetuses evaluated for external malformations	45	45	45	33	37	45	110
No of malformed fetuses	2	1	2	4	14	18	1
Cutaneous cyst	—	—	—	—	8	3	—
Microphthalmia	—	—	—	—	8	—	—
Nasal septum lesion	—	—	—	—	7	—	—
Cleft palate	1	—	1	3	6	9	1
Hydrocephaly	—	1	—	—	7	3	—
Gastroschisis	—	—	—	1	3	2	—
Subcutaneous hemorrhage	—	1	—	—	1	3	—

Lab animals

Female BALB/c mice within a weight range of 25–30 g were used. The mice were taken from the Pharmaceutical Sciences Research Center at Bu Ali Institute, Mashhad University of Medical Sciences, and kept under conditions of 12-hour darkness and 12-hour light, 23 ± 2 degrees centigrade and appropriate moisture. One male mouse was used for mating with five female mice. Three hours after mating, female mice were evaluated for vaginal plaque and those having the plaque were considered pregnant on day zero of gestation (GD₀).^{8–10} Each group consisted of 15 mice.

Extraction of fetuses and staining

The experimental protocol for evaluating teratogenicity was based on the ICH guidelines.^{11,12} The female mice were administered chloroform on day 18 of gestation; the abdomen was incised over the uterus, the uterus and fallopian tubes were extracted and weighed. Subsequently, the fetuses were sectioned out of amniotic sac. The total number of fetuses, the number of live and dead fetuses, their weight (± 1 mg), crown-rump (± 0.1 mm) and the number of early and late resorption were determined.¹³ After their extraction from the amniotic sac, the fetuses were deposited in acetone for 24 hours, then their skin was scraped off and they were submerged in stain solution including Alcian Blue (Sigma, UK), Alizarin Red-S (Sigma, UK), ethanol and acetic acid. Then they were stained using microwave radiation.¹⁴ Transparency was achieved by a solution containing 1% potassium hydroxide, 95% ethanol and glycerin for 24 hours. Subsequently, the fetuses were evaluated for structural abnormalities including anatomical,

morphological, skeletal and osseous properties using a stereomicroscope with scaled face (Zeiss).^{8,15,16}

Data analysis

The data were reported as mean $\pm$ standard deviation for each group. Analysis of variance (ANOVA) test followed by Turkey-Kramer test were used to evaluate the difference of means among more than two groups (Instat software, Ver 3). *p* Values of equal to or smaller than 0.05 were considered significant.

Results

The weight changes of pregnant mice from day 0 through 18 for different doses of HESA-A are shown in Figure 1. Oral administration of HESA-A started on day 6 of gestation. The results indicate that in the control group and in doses less than 400 mg/kg, weight gain progressed without interruption from GD₀ through GD₁₈. Those receiving 400 mg/kg had slower weight gain only on day 14. Those mice receiving 800 mg/kg had evident slowing in weight gain from day 8 onwards. Despite discontinuation of the compound on day 14, this disturbance continued through day 18.

Table 1 records the evaluated variables including number of female mice for mating, number of pregnant mice, fertility index, number of pregnant mice in study, weight of uterus, number of corpora lutea, number of implantations, number of pre-implantation loss, number of resorptions, average number of fetuses per mouse, total number of fetuses per group, average number of viable fetuses per pregnant mouse, weight of fetus and height of fetus. As can be seen, the weight of uterus in those mice receiving 400 and 800 mg/kg show a significant decline compared to those receiving normal saline (*p* < 0.01). Pre-implantation loss was

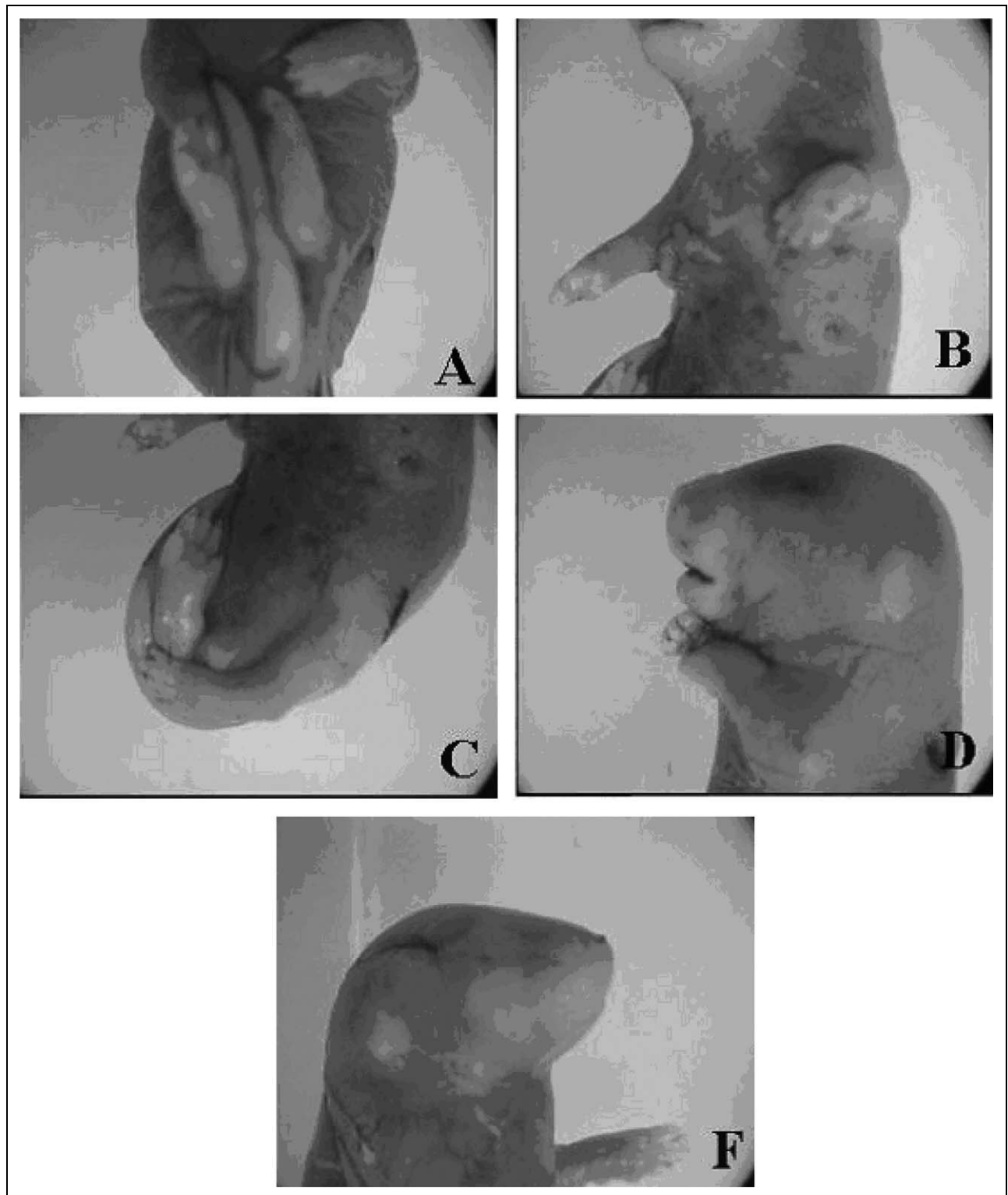


Figure 2. External abnormalities in fetuses receiving HESA-A 400 mg/kg (A: short leg, D: hydrocephaly, E: twisted extremities) and 800 mg/kg (B: short arm, C: twisted vertebral column, F: short neck).

significantly different from the normal saline group only at the highest dose ($p < 0.05$). The number of resorptions is significantly higher in those receiving

400 and 800 mg/kg doses in comparison to normal saline ($p < 0.01$). The number of remaining fetuses sectioned out of the uterus of pregnant mice on day 18

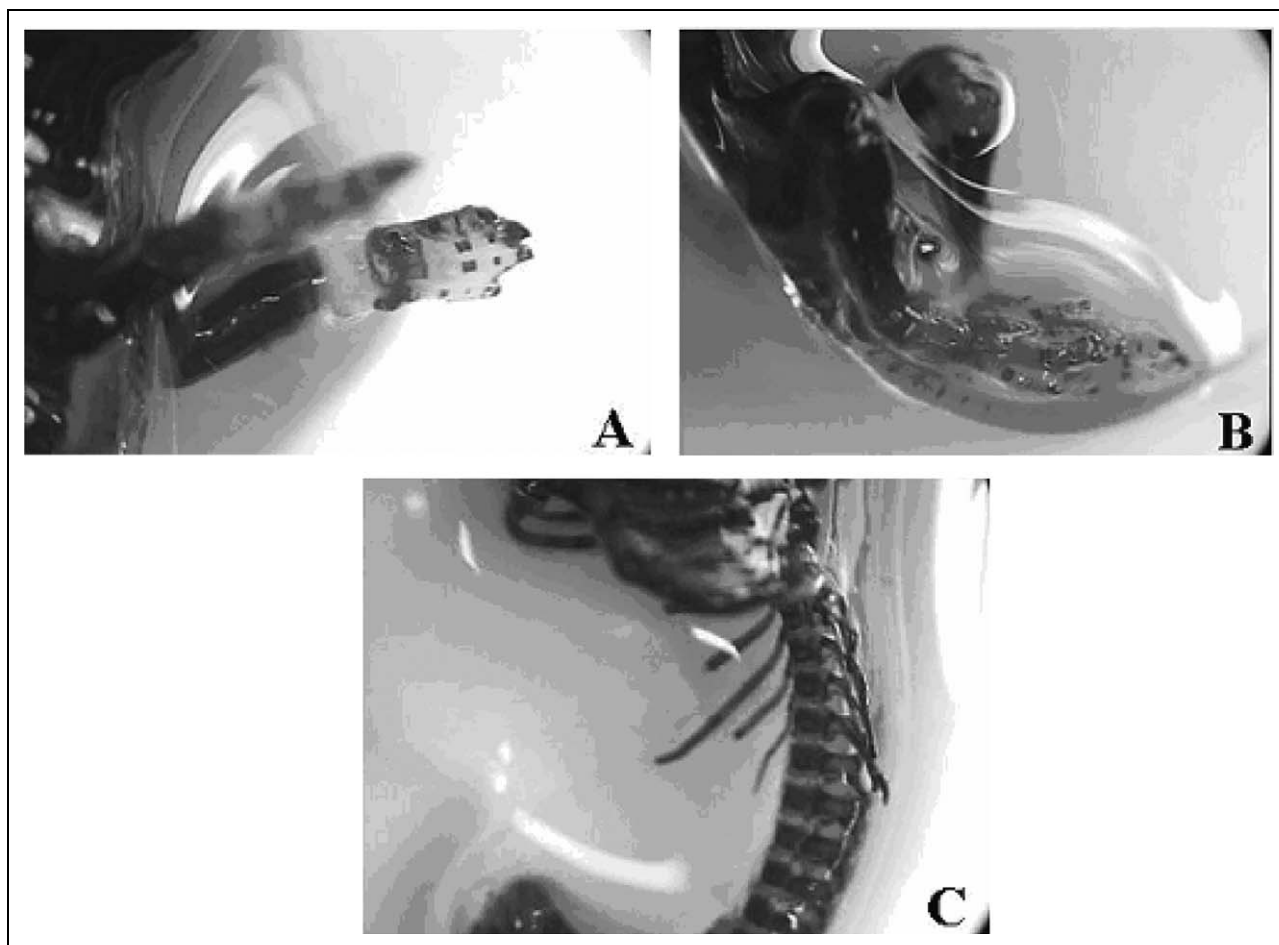


Figure 3. Skeletal abnormalities of fetuses receiving HESA-A 400 mg/kg (C: costal process) and 800 mg/kg (A: agenesia of hand cartilages, B: agenesia of foot cartilages).

was significantly lower in the 400 and 800 mg/kg groups compared to the normal saline group ($p < 0.01$). The number of viable fetuses is also lower in three treatment groups (200, 400 and 800 mg/kg) in comparison to normal saline group. The weight of fetuses in those mice receiving 400 and 800 mg/kg is significantly lower compared to normal saline ($p < 0.01$). Moreover, the height of fetuses in the 400 and 800 mg/kg groups is significantly lower in comparison to normal saline group ($p < 0.05$).

Table 2 summarizes the results of fetuses evaluated for external abnormalities. The external abnormalities have been depicted for different doses of HESA-A which indicates that higher doses result in more abnormalities. Malformations in the negative control group are within acceptable historical ranges for BALB/c mice.

Figure 2A shows a fetus receiving HESA-A in 400 mg/kg doses, with evident short legs. Figure 2B

shows a fetus receiving HESA-A in 800 mg/kg doses with short arms. Figure 2C shows the twisted vertebral column in a fetus receiving HESA-A in 800 mg/kg doses. Figure 2D belongs to a fetus receiving HESA-A in 400 mg/kg doses with evident hydrocephaly. Figure 2E shows twisted extremities in a fetus receiving HESA-A in 400 mg/kg doses. Figure 2F shows a fetus receiving HESA-A in 800 mg/kg doses, which has been afflicted with short neck.

The skeletal abnormalities in fetuses receiving various doses of HESA-A are summarized in Table 3, which include different abnormalities such as short ribs and lack of ossification of various bones. These abnormalities were dose-dependent as well. The skeletal abnormalities are depicted in Figure 3. Figure 3A shows a fetus receiving HESA-A in 800 mg/kg doses, with agenesia of hand cartilages. Figure 3B belongs to a fetus receiving HESA-A in 800 mg/kg doses, with agenesia of foot cartilages. Figure 3C depicts a fetus

Table 3. Evaluation of skeletal findings in fetuses receiving HESA-A in different doses

Skeletal findings	HESA-A (mg/kg)					Cyclophosphamide (15 mg/kg)	Normal saline
	50	100	200	400	800		
No fetuses evaluated for skeletal malformations	69	48	39	33	32	41	77
No of malformed fetuses	1	2	2	8	12	18	1
Reduced ossification	–	1	–	5	9	13	–
Fused sternum	1	–	–	4	5	–	–
Shortness of long bones	–	2	1	3	9	11	–
Short ribs	–	1	1	2	11	11	–
Costal processes	–	1	–	6	9	8	–
Lack of ossification of tail vertebrae	–	–	–	–	1	6	–
Abnormal twisting of extremities	–	–	1	1	–	–	–
Short arms of legs	–	–	–	3	7	–	4

receiving HESA-A in 400 mg doses with evident costal processes. All above evaluations used cyclophosphamide as positive control.

Discussion

As can be seen, the weight change of pregnant mice is normal from day zero through 6, but the weight change of the group receiving HESA-A in high doses progresses differently compared to the normal saline group. Although HESA-A administration was discontinued on day 14, the weight change pattern remains significantly different from the normal saline group, indicating the chronic and long-term impact of the compound on the nutritional status of mice. On the other hand, low and intermediate doses of HESA-A do not have considerable effects on weight gain of pregnant mice, a finding that indicates the substance's lack of general toxicity for pregnant mice.

The weight of uterus of the mice receiving HESA-A in 400 mg/kg doses is significantly lower compared to the normal saline group, presumably due to the intra-uterine growth retardation caused by the substance. Studies conducted on teratogenic compounds such as cobalt sulfate indicate that placental hypoperfusion and consequently fetal hypoperfusion and maternal or fetal metabolic disturbances are the cause of growth retardation in uterus.^{17,18} Therefore, the impact of HESA-A in causing intra-uterine growth retardation might be mediated through fetal hypoperfusion and fetal or maternal metabolic disturbance. The dose-dependent increase in number of resorption indicates that the compound increases fetal mortality. The decline in fetal weight and height and increase in number of dead fetuses corroborates the fact that HESA-A causes embryotoxicity in a dose-dependent fashion.

As mentioned earlier, HESA-A increases skeletal and external abnormalities in a dose-dependent mode. As for the external abnormalities such as hydrocephaly, gastroschisis and subcutaneous hemorrhage, it is likely that the organic constituents in HESA-A are to blame, as different studies confirm the strong teratogenicity of organic compounds (such as formaldehyde and benzene).¹⁹⁻²¹ The skeletal abnormalities observed in HESA-A receiving fetuses most frequently include lack of ossification of different bones, probably through replacement of calcium with heavy metals in HESA-A. Previous studies confirm that heavy metals such as aluminum and chromium cause osseous abnormalities by the same mechanism.^{22,23} No considerable abnormality of mice was observed in 200 mg/kg dose – 4 times the dose used in clinics. The findings indicate that at the doses commonly used in clinics, HESA-A has no toxic impact on mouse fetus.²⁴ Although in one study it was reported that apoptosis has a role in HESA-A teratogenicity,²⁵ more studies are needed to clarify this effect. Most anticancer agents exert their teratogenic effects at their clinical doses, whereas HESA-A has not been observed to follow this pattern and its potential for embryotoxicity only occurs at high doses. Extensive studies are required to clarify the teratogenic mechanism of HESA-A. Another important area that needs to be studied is pharmacokinetic parameters of HESA-A to evaluate rate and extent of absorption, distribution, metabolism and elimination that would greatly help interpretation of above results.

Conclusion

According to the findings of this study, HESA-A has no impact on growth parameters, external anatomical

structures, and cartilage and bone development of mouse fetus, even in doses higher than therapeutical doses. However, at very high doses of the compound, different adverse effects are observed. It should be emphasized that these adverse effects occur at doses considerably higher than those used in clinics. In any case, administration of drugs during pregnancy must be done with extra care and all benefits and disadvantages must be evaluated thoroughly.

Acknowledgement

We would like to thank Farzan Institute for Research and Technology for their help in manuscript preparation.

Funding

This study was supported by the Research Council of Mashhad University of Medical Sciences for their financial support.

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